Research Update

Folic acid supplementation has sustained benefits beyond the first trimester by improving methylation capacity

Article highlights

- Periconceptional folic acid supplementation and food fortification is known to have a protective effect against neural tube defects, however the effects of supplementation after the first trimester are not fully known.
- In a randomised clinical trial, folic acid supplementation (400µg/d) during the second and third trimesters not only led to significant protection against folate depletion in mothers and offspring, but also improved cognitive performance in children during the first decade of life.
- Continued maternal supplementation with folic acid through trimesters 2 and 3 of pregnancy resulted in significant changes in DNA methylation of candidate genes related to brain development in newborns; highlighting the association between maternal folate status, DNA methylation and offspring neurodevelopment.

Periconceptional folic acid has been demonstrated to have a protective effect in the prevention of neural tube defects (NTD), leading to global recommendations for folic acid supplementation before and during early pregnancy, and to the introduction of mandatory folic acid fortification in many countries.[1] In Australia, the incidence of neural tube defects has decreased from 10.2 per 10,000 births pre-mandatory fortification (2009) to 8.7 per 10,000 births post-mandatory fortification, which is a 14.4% decrease.[2] After the 12th week of pregnancy there are no official guidelines for folic acid intake, however, emerging epidemiological evidence suggests that maternal folate supplementation throughout the entire pregnancy may have other beneficial roles in the health of offspring, including neurodevelopment and cognitive performance in the first decade of life.[3] These findings have prompted researchers to further study the effects of continued folic acid supplementation during pregnancy.

In the first randomised controlled trial (RCT) of folic acid supplementation specifically in the second and third trimesters (Folic Acid Supplementation in the Second and Third Trimesters (FASSTT) trial), researchers aimed to investigate the effects of continued folic acid supplementation on maternal folate and homocysteine responses and related effects in newborns.[4] In this double-blind, randomised, placebo-controlled trial supplementation, 400µg/d of folic acid in the second and third trimesters prevented significant changes in maternal serum folate and plasma homocysteine and significantly increased red blood cell folate concentrations between the 14th gestational week (GW) (pre-intervention) and the 36th gestational week. In the placebo group, concentrations of maternal serum and red blood cell folate were significantly lower, and plasma homocysteine concentrations were significantly higher at the end of treatment compared to the folic acid group. Analysis of cord blood showed significantly higher serum and red blood cell folate concentrations in infants of mothers supplemented with folic acid.[4]

Follow up of mother-child pairs from the original FASSTT trial was conducted over a decade to assess the effect of folic acid supplementation during the second and third trimesters of pregnancy on cognitive development in children. The results showed that children of mothers supplemented with folic acid throughout pregnancy performed better in the cognitive domain at age 3 years,[5] the verbal domain at 7 years[6] and processing speed domain at 11 years.[7] These results indicate that folic acid supplementation throughout pregnancy may impact neurodevelopment and cognitive performance in the first decade of life.

The mechanisms by which folic acid supplementation prevents NTDs and influences neurodevelopment are poorly understood but could involve epigenetic changes.[1,8] Folate is a critical component in the one-carbon metabolism pathway providing methyl groups for a range of biochemical reactions, including methylation of DNA.[8] DNA methylation is an important epigenetic determinant of gene expression, and changes in methylation have been associated with multiple diseases, such as NTD, and impaired early life development.[8,9] Periconceptional maternal folate levels may alter methylation patterns established in utero that are vital for foetal development, which could impact later health and developmental outcomes in the offspring.[9]

There has been growing interest in the importance of maternal folate status, DNA methylation and offspring neurodevelopment. To better understand the epigenetic mechanisms linking maternal folate and neurodevelopment researchers used available blood cord samples from the FASSTT trial in pregnancy to study 9 candidate loci known to be regulated by methylation.[8] This study showed significant folate-related changes in DNA methylation of candidate genes related to brain development such as insulin-like growth factor 2 (IGF2) and brain-derived neurotrophic factor (BDNF) in the newborn of mothers who received 400µ/d folic acid compared with placebo in the second and third trimesters.[8]

Emerging evidence from the FASSTT study demonstrates that folic acid supplementation during the second and third trimesters not only led to significant protection against folate depletion in mothers and offspring, but also improved cognitive performance in children during the first decade of life. Emerging evidence suggests that folic acid supplementation may be beneficial for foetal brain development in later pregnancy via its effect on DNA methylation of specific genes relating to the brain.

References

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